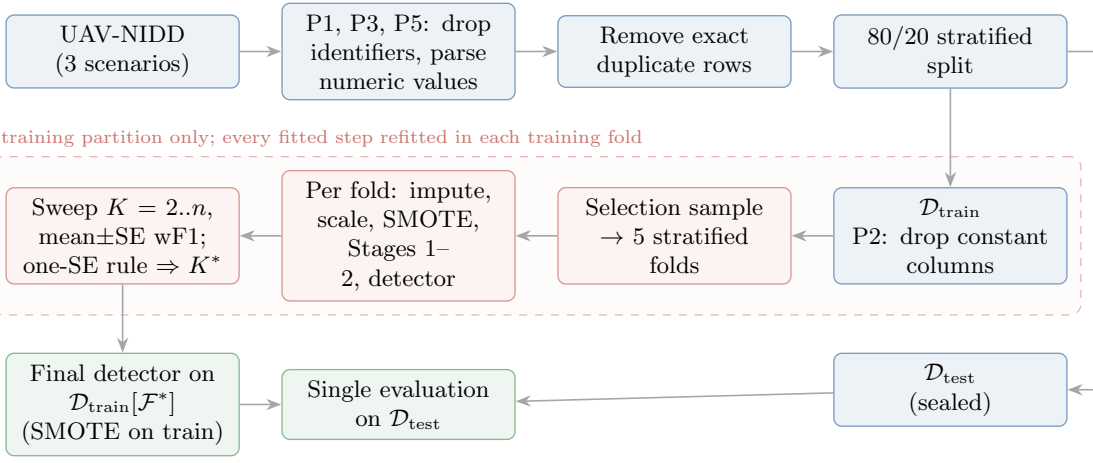

UAV-NIDD
(3 scenarios)
P1, P3, P5: drop identifiers, parse numeric values
Remove exact duplicate rows
80/20 stratified split
training partition only; every fitted step refitted in each training fold
$\mathcal{D}_{\text{train}}$
P2: drop constant columns
Selection sample → 5 stratified folds
Per fold: impute, scale, SMOTE, Stages 1–2, detector
Sweep $K = 2..n$, mean±SE wF1; one-SE rule ⇒ K^*
Final detector on $\mathcal{D}_{\text{train}}[\mathcal{F}^*]$ (SMOTE on train)
Single evaluation on $\mathcal{D}_{\text{test}}$
$\mathcal{D}_{\text{test}}$
(sealed)